UCS 503 Software Engineering Project Proposal

CampusCycle

Smart Campus Bicycle Sharing and Mobility Management System

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Project Proposal

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Chapter 1

Introduction

1.1 Background and Context

CampusCycle is a smart bicycle-sharing platform designed specifically for university campuses. The system allows students to locate, reserve, unlock, use and return shared bicycles through a web application. Administrators can manage bicycles, stations, rentals, maintenance and usage analytics.

The project extends beyond basic rental functionality by incorporating QR-based bicycle identification, geofencing, maintenance reporting, usage analytics and demand prediction. These features aim to make short-distance transportation within a large campus more convenient, affordable, sustainable and manageable.

Shared-bicycle schemes are well established as a solution to the last-mile problem and have been deployed in several hundred cities worldwide [9]. The research literature on these schemes consistently identifies two factors as decisive for whether a scheme is adopted or abandoned: reliable information about where bicycles are available, and the condition of the fleet [3]. Both are information problems rather than hardware problems, which is what makes a campus scheme a suitable software engineering project.

1.1.1 Problem Statement

Large university campuses often require students to travel considerable distances between hostels, academic blocks, libraries, cafeterias, sports facilities and administrative buildings. Although bicycles can solve this last-mile mobility problem, a shared bicycle system requires efficient tracking and management.

A conventional rental system does not adequately address problems such as uneven bicycle distribution, unauthorised use outside the campus, maintenance issues, and difficulty in predicting where bicycles will be needed. Studies of operating schemes report that redistributing bicycles between stations is one of the largest recurring operational costs [2], and that cost is driven by decisions taken without reliable demand information.

Therefore, the proposed system aims to develop a centralised smart campus bicycle-sharing platform that manages the bicycle fleet while using location and historical usage data to improve availability and operational efficiency.

1.2 Project Objectives

The main objectives of CampusCycle are:

1. Provide students with a simple platform to find and rent campus bicycles.
2. Implement QR-based identification for individual bicycles.
3. Display bicycle and docking-station availability on an interactive campus map.
4. Maintain complete rental and return records.
5. Use geofencing to detect bicycles leaving the permitted campus area.
6. Allow students to report bicycle faults and maintenance problems.
7. Provide administrators with fleet, usage and maintenance analytics.
8. Analyse historical rental data to predict bicycle demand at different stations.
9. Recommend redistribution of bicycles between stations based on predicted demand.

Chapter 2

Proposed System

The system will contain two major interfaces: a student application and an administrator dashboard.

2.1 Student Application

Students will be able to:

- Register and log in using their university account.
- View nearby bicycle stations and available bicycles.
- Scan a QR code to identify a bicycle.
- Reserve and start a bicycle rental.
- View the active ride and rental duration.
- Return the bicycle at an approved station.
- View previous rides and distance travelled.
- Report mechanical or safety issues.
- View personal mobility statistics and sustainability contributions.

2.2 Administrator Dashboard

Administrators will be able to:

- Add, remove and update bicycles and stations.
- Monitor active rentals.
- Track bicycle availability and locations.
- View maintenance reports and bicycle status.
- Detect geofence violations.
- Analyse station-wise and time-wise usage.
- View predicted demand for upcoming periods.
- Receive recommendations for bicycle redistribution.

2.3 Key Features

2.3.1 QR-Based Bicycle Identification

Each bicycle will have a unique QR code. Scanning the code allows the backend to identify the bicycle and verify whether it is available, reserved, currently rented or under maintenance.

2.3.2 Campus Geofencing

A virtual boundary will be defined around the campus. In an extended implementation with GPS-enabled hardware, the system can detect when a bicycle crosses this boundary and generate an alert for the administrator. For the software prototype, location data can be simulated or obtained from a mobile device.

2.3.3 Interactive Campus Map

The application will display stations and their current availability. Stations can be represented using status indicators such as available, low availability and unavailable.

2.3.4 Maintenance Management

Students can report issues such as damaged brakes, flat tyres or faulty lights. Repeated reports can increase the priority of a maintenance request, and bicycles marked unsafe can automatically become unavailable for new rentals.

2.3.5 Demand Prediction

Historical rental information will be analysed using machine-learning techniques. Features may include station, time of day, day of week, previous demand and number of active rentals. A regression or tree-based model will predict future demand.

2.3.6 Smart Redistribution

Predicted demand will be compared with current station capacity. The system can recommend moving bicycles from stations with excess supply to stations where demand is expected to be high.

Chapter 3

System Architecture and Technology

3.1 Proposed Architecture

The proposed architecture consists of:

1. **Frontend:** React.js [5] with Tailwind CSS for the student and administrator interfaces.
2. **Backend:** Node.js and Express.js [6] for authentication, rentals, bicycle management, maintenance and REST APIs.
3. **Database:** MySQL for users, bicycles, stations, rentals, locations, maintenance records and analytics data.
4. **Machine Learning Module:** Python with Scikit-learn [8] for demand prediction and redistribution recommendations.
5. **Mapping:** Leaflet [1] with OpenStreetMap [7], or another suitable mapping service, for the campus map.
6. **Optional Hardware Extension:** ESP32 with GPS and a lock mechanism for demonstrating real-time bicycle tracking.

3.2 Technology Stack

Component	Technology
Frontend	React.js, Tailwind CSS
Backend	Node.js, Express.js
Database	MySQL
Machine Learning	Python, Scikit-learn, Pandas
Maps	Leaflet / OpenStreetMap
Authentication	JWT [4] / University Authentication
Version Control	Git and GitHub
Optional Hardware	ESP32, GPS module, electronic lock

Table 3.1: Proposed technology stack.

3.3 Project Scope

The minimum viable product will include authentication, bicycle and station management, QR-based identification, rental and return workflows, campus map visualisation, maintenance reporting, and an administrator dashboard.

Advanced features such as real-time GPS tracking, electronic locks, demand prediction and smart redistribution will be implemented as higher-level modules depending on available development time and hardware availability.

Chapter 4

Expected Outcome and Conclusion

4.1 Expected Outcome

At the end of the project, a working prototype of CampusCycle will allow a student to discover an available bicycle, scan its QR code, start a rental, view the ride status and return the bicycle at an approved station. Administrators will have a dashboard for fleet and maintenance management.

The system will additionally demonstrate analytics and a machine-learning model capable of predicting station-level bicycle demand. The prototype will provide a foundation that could later be extended to a real campus deployment with GPS trackers, electronic locks, payment integration and mobile applications.

4.2 Conclusion

CampusCycle combines web development, databases, geospatial concepts, machine learning and optional IoT hardware into a practical campus mobility solution. The project addresses a real transportation problem while remaining feasible as a two-person semester project. Its modular design also allows the basic bicycle-sharing system to be developed first and advanced intelligence and hardware features to be added incrementally.

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